1 Mathematical Framework

1.1 Model (Prototype)

The model depends on hyperparameters

$$\sigma_1, \dots, \sigma_K, \lambda_1^{(0)}, \dots, \lambda_K^{(0)}, \lambda_1^{(1)}, \dots, \lambda_K^{(1)}, n_1, \dots, n_K, c_1, \dots, c_K$$

$$1, 2, \dots, K \text{ arms}, \ 1, 2, \dots, T \text{ Time Horizon}$$

$$Y_{i,t}^{(\sigma_i)} \sim N(Z_{i,t}, \sigma_i^2) \text{ Reward of arm } k \text{ at time } t$$

$$Z_{i,t} = \mathbb{E}(Y_{i,t}^{(\sigma)}) \in \mathcal{Z}$$

$$\mathcal{Z} = \left\{ Z \in \{0,1\}^{K \times T} : \sum_{t=1}^{T} Z_{i,t} = n_i, \ \frac{1}{T} \sum_{t=1}^{T-1} Z_{i,t} Z_{i,t+1} \geq c_i \right\}$$

$$a_t \subseteq \{1, \dots, K\} \text{ active arms at time } t$$

$$\text{Let } 1_{i,t} = 1(i \in a_t), \text{ Let } H_i^{(0)}(t) = \min\{s \geq 1 : 1_{i,t+s} \neq 1_{i,t}, \ 1_{i,t} = 0\} \text{ Sleeping holding time}$$

$$\text{Let } H_i^{(1)}(t) = \min\{s \geq 1 : 1_{i,t+s} \neq 1_{i,t}, \ 1_{i,t} = 1\} \text{ Awake holding time}$$

$$H_i^{(j)}(t) \sim \text{Geom}(\lambda_i^{(j)}), \ i.e \ \mathbb{P}(H_i^{(j)}(t) = s) = (1 - \lambda_i^{(j)})^{s-1} \lambda_i^{(j)}, \ \ s \geq 1$$

Let X_t be the decisions of the algorithm $f \in \mathcal{F}_t \ s.t \ \mathcal{F}_t = \left\{ f : \{Y_{X_s,s}^{(\sigma_{X_s})}, X_s\}_{s=1}^t \to \{1, \dots, K\} \right\} \ \forall \ t \in \{1, \dots, T\}$

We now formalize the regret:

$$R_T = \mathbb{E}\left(\sum_{t=1}^{T} Y_{i^*,t}^{(\sigma_{i^*})} 1_{i^*,t} - \sum_{t=1}^{T} Y_{X_t,t}^{(\sigma_{X_t})} \right) = \sum_{t=1}^{T} Z_{i^*,t} 1_{i^*,t} - \sum_{t=1}^{T} Z_{X_t,t}$$

$$i^* = \arg\max_{i \in [K]} \sum_{t=1}^{T} Z_{i,t} 1_{i,t}$$

1.2

2 Supplementary Material for Code

- `src/hybrid.py <class object> gauss_hybrid_model` : Refer to A1

3 A1

Algorithm 1 Generate Adversarial Sequence

Require: Horizon T, Autocorrelation Parameter c, Observation Ratio ρ

Ensure: Sequence $X_1, X_2, \ldots, X_T$ where $X_i \in \{0,1\}$, $\frac{1}{T}\sum_{t=1}^{T} X_t = \rho$

$X_1 \sim Bernoulli(\rho)$

for $t = 2, \ldots, T$ **do**

  $\hat{p}_t \leftarrow \frac{1}{t-1}\sum_{s=1}^{t-1} X_s$ ▷ *Running empirical mean*

  $e_t \leftarrow \hat{p}_t - \rho$ ▷ *Error from target*

  if $X_{t-1} = 1$ **then**

    $p_{\text{stay}} \leftarrow \frac{1+c}{2} - e_t$

  else

    $p_{\text{stay}} \leftarrow \frac{1+c}{2} + e_t$

  $p_{\text{stay}} \leftarrow \text{clip}(p_{\text{stay}},\ 0,\ 1)$

  stay $\sim$ Bernoulli(p_{stay})

  if stay **then**

    $X_t \leftarrow X_{t-1}$

  else

    $X_t \leftarrow 1 - X_{t-1}$

return $X_1, \ldots, X_T$